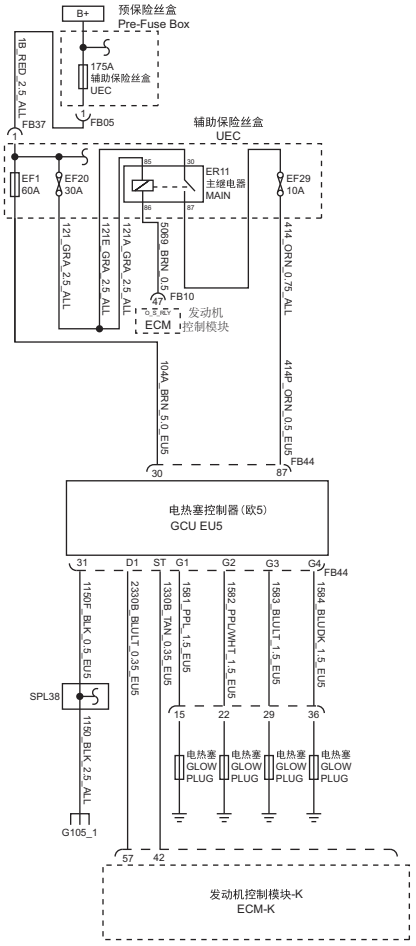
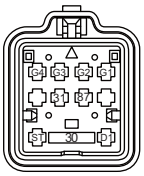
P037D、P0381、P064C、 P0671、P0672、 P0673、P0674

Description of fault diagnosis codes
P037D-11：Short circuit to ground error for Glow plug Low Voltage System
P037D-12：Short circuit to battery error for Glow plug Low Voltage System
P037D-4B：Over temperature error on ECU powerstage for Glow plug Low Voltage System
P0381-F4：No load error for Glow plug Low Voltage System
P064C-09：DFC for glow module error in GCU-T
P064C-41：DFC for coding error when selected coding is not matching for glow plug
P064C-F1：DFC for faulty diagnostic data transmission or protocol error for glow plug
P064C-F2：DFC for coding error when different coding words were received in a coding cycle for glow plug
P064C-F3：DFC for T30 missing in GCU-T: If no voltage is present at the T30 input when GCU is activated GCU-T
P0671-01：DFC for short circuit Glow Plug 1 powered by GCU-T
P0671-07： DFC for failure Glow Plug 1 powered by GCU-T
P0672-01：DFC for short circuit Glow Plug 4 powered by GCU-T
P0672-07：DFC for failure Glow Plug 4 powered by GCU-T
P0673-01：DFC for short circuit Glow Plug 2 powered by GCU-T
P0673-07：DFC for failure Glow Plug 2 powered by GCU-T
P0674-01：DFC for short circuit Glow Plug 3 powered by GCU-T
P0674-07：DFC for failure Glow Plug 3 powered by GCU-T
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Glow plug controller fault • Engine control module fault.
Description of circuit diagram
The preheating plug controller is connected to the engine control module via wire, the main relay provides 12v voltage to the preheating plug controller, and the engine control module controls the power supply of preheating plug through receiving the feedback signal of preheating plug controller and sending control signal to the preheating plug controller.

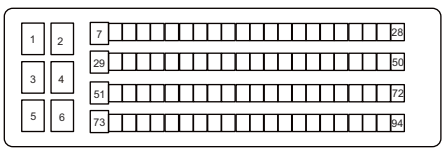
Circuit diagram



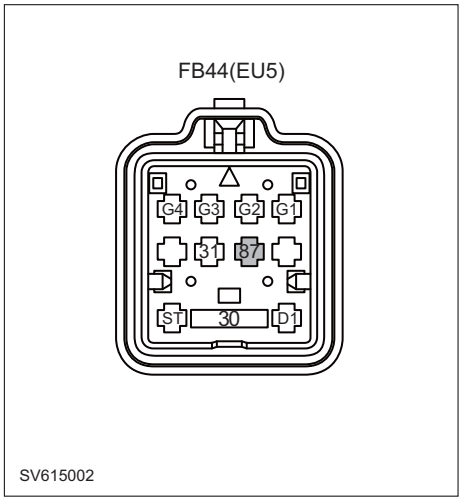
FB44(EU5)

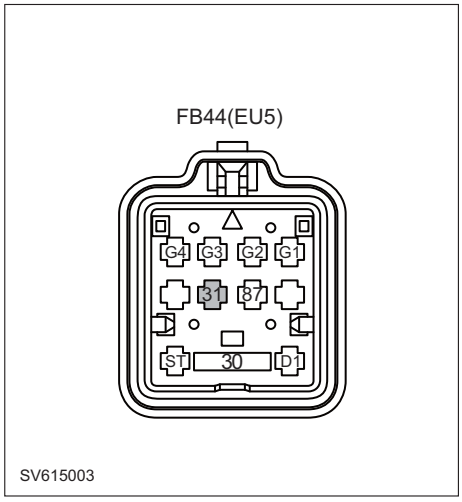


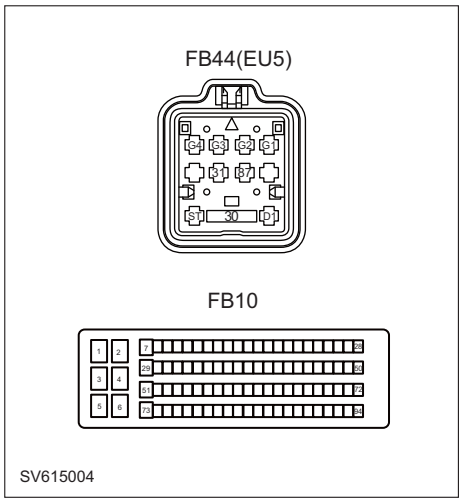
FB10



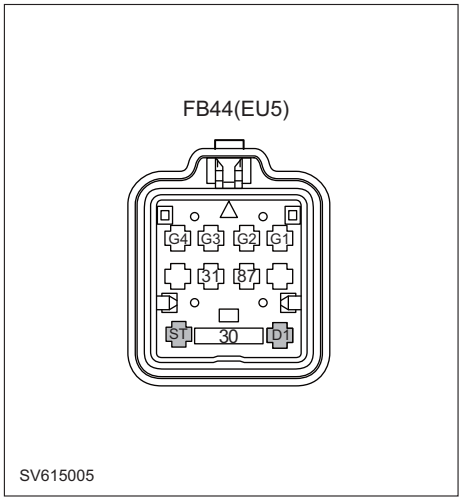
SV615001

Troubleshooting procedures	
Step 1	General inspection.
	1 Check the fuse EF29 for damage. 2 Check battery capacity. 3 Check the engine control module and harness plug of glow plug controller for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not.
	No → Repair or replace the faulted parts.
No ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault except for P037D, P0381,P064C, P0671,P0672,P0673,P0674?
	Yes → Reference: DTC Summary List (EU5).
No ↓	
Step 3	Check the power supply circuit of glow plug controller.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of glow plug controller. 3 Turn the ignition switch to ON. 4 Check the voltage between the harness plug of glow plug controller and ground. Measure circuit: the voltage between FB44 harness plug terminal 87 and ground. **Standard value: 10~14V** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 4	Check the ground circuit of glow plug controller.
	1 Turn the ignition switch to OFF. 2 Check the resistance between the harness plug of glow plug controller and ground. Measure circuit: the resistance between FB44 harness plug terminal 31 and ground. **Standard value: ≈ 0Ω** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 5	Check the control circuit of glow plug controller (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Measure the resistance of harness plug between engine control module and glow plug controller. Measure circuit: the resistance between FB44 plug terminal ST and FB10 plug terminal 42. Measure circuit: the resistance between FB44 plug terminal D1 and FB10 plug terminal 57. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the control circuit of glow plug controller (short circuit).



1 Measure the resistance between the harness plug of glow plug controller and ground.

Measure circuit: the resistance between FB44 plug terminal ST and ground.

Measure circuit: the resistance between FB44 plug terminal D1 and ground.

Standard value: ∞

2 Measure the voltage between the harness plug of glow plug controller and ground.

Measure circuit: the voltage between FB44 plug terminal ST and ground.

Measure circuit: the voltage between FB44 plug terminal D1 and ground.

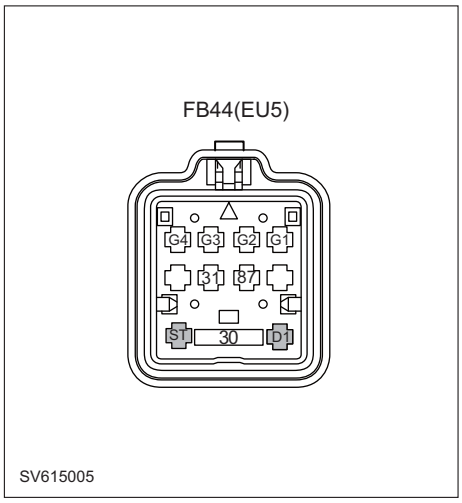
Standard value: ≈ 0V

3 Check whether the result is normal?

No → Repair or replace the faulted parts.

Yes ↓

Step 7	Check the glow plug controller (components) .



1 Connect the harness plug of engine control module.

2 Turn the ignition switch to ON position.

3 Measure the voltage value between the harness plug of glow plug controller and ground.

Measure circuit: the voltage between FB44 plug terminal ST and ground.

Measure the circuit: the voltage between FB44 plug terminal D1 and ground.

Standard value: 10.2~11.2V

4 Check whether the result is normal?

No → Replace engine control module.

Yes ↓

Step 8	Replace the glow plug controller.

1 Replace the glow plug controller.

Next ↓

Step 9	Complete the check and confirm whether the system works normally nor not.